

Plasmonic lattice lasers

Francisco Freire-Fernández^{1,2}, Sang-Min Park^{1,2}, Max J. H. Tan¹ & Teri W. Odom¹✉

Abstract

Plasmonic lattice lasers offer a promising alternative to compact sources such as vertical-cavity surface-emitting lasers. These lasers have an open-cavity design consisting of periodic lattices of metallic nanoparticles that facilitate integration with both liquid-state and solid-state gain nanomaterials. Recent advances have enabled real-time control over lasing wavelength, tunable multimodal lasing, and design of complex polarization and intensity profiles. In this Review, we summarize key developments in plasmonic lattice lasers over the past 5 years, with a focus on unconventional lattice cavities and how they can facilitate tailored lasing characteristics. We discuss strategies for realizing multicolour and multidirectional emission, the advantages of different gain materials and the challenges of reducing lasing thresholds. Although substantial progress has been made, open questions regarding fabrication precision, threshold engineering and the realization of electrically driven plasmonic lasers remain. Plasmonic lattice lasers are poised to play a critical part in next-generation technologies for optical communication, sensing and quantum applications.

Sections

Introduction

Plasmonic lattice laser components

Recent progress in plasmonic lattice lasers

Future perspectives and challenges

¹Department of Chemistry, Northwestern University, Evanston, IL, USA. ²These authors contributed equally: Francisco Freire-Fernández, Sang-Min Park. ✉e-mail: todom@northwestern.edu

Introduction

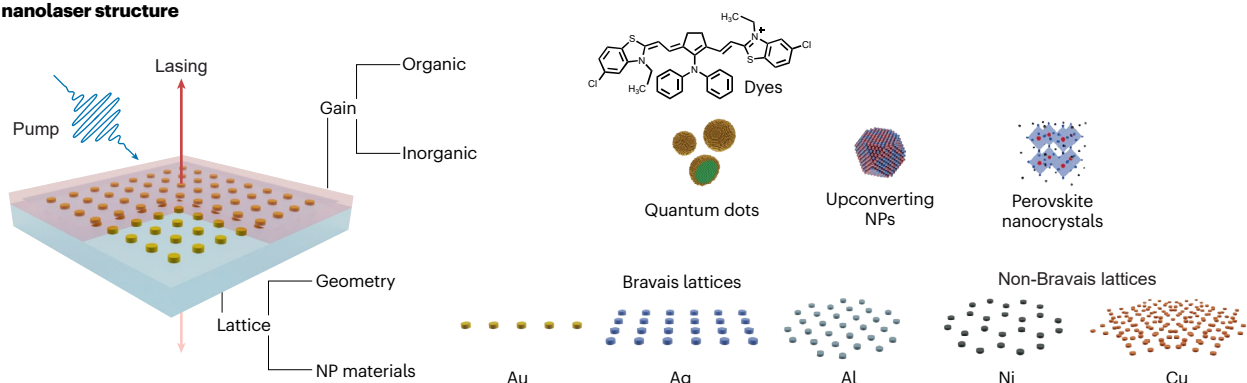
The demonstration of amplified stimulated emission of microwave radiation in the 1950s set in motion research that resulted in the development of the laser¹. Starting with their use in surgery in 1961 (ref. 2), lasers rapidly transformed other fields. Continuous advances in materials and miniaturization resulted in applications such as bar-code scanners and a lightwave communications system under downtown Chicago in the 1970s^{3,4}. Currently, vertical-cavity surface-emitting lasers (VCSELs) are the industry standard for consumer electronics and optical communication⁵. VCSELs consist of two mirrors that form a Fabry–Perot cavity surrounding a gain material, typically GaAs-based quantum wells⁵. The distributed Bragg reflector mirrors support optical modes with high quality factor, which are important for light confinement, optical feedback, and emission normal to the cavity. Despite their use in many technologies, VCSELs have limited flexibility to produce laser characteristics such as beam polarization, intensity profile and directionality, which are increasingly desirable for emerging technologies that require encryption and high spatial resolution from a single laser source^{6–9}. Hence, alternative cavity architectures based on 2D periodic structures such as photonic crystals and plasmonic nanoparticle lattices have recently been pursued^{10,11}.

Dielectric photonic crystals are diffraction-limited and confine light within the material^{10,12}. Photonic crystal lasers rely on solid-state, high-refractive-index, semiconducting gain materials^{13–17}. By contrast, plasmonic nanoparticle lattices feature an open-cavity architecture that enables direct integration of both liquid-state and solid-state gain materials^{10,18,19} (Fig. 1a). In 2004, theory predicted how periodic

metal nanoparticle arrays could effectively reduce optical losses through diffractive coupling^{20,21} (Fig. 1b), offering a preview of their potential as lasing cavities that would be more advantageous than those based on single plasmonic nanoparticles, nanowires and films^{22–26}. These calculations were validated experimentally in 2008 by reports of plasmon modes with remarkably narrow linewidths^{20,27}. In 2013, plasmonic lattice lasers were realized using these optical cavity modes combined with dye-doped polymer matrices; modelling of the lasing action was performed using a semi-quantum electromagnetic approach²⁸.

Plasmonic lattice lasers have historically been fabricated out of noble metals (Au, Ag) and with Bravais lattice symmetries. Work has then expanded to include non-noble metals^{29–32} and non-Bravais lattices^{33–35} and to integrate a range of organic and inorganic gain nanomaterials (Fig. 1a). Key accomplishments in plasmonic lattice lasers based on advances in their constituent components and theory are highlighted in Fig. 1b. Real-time tuning of lasing wavelengths was possible by controlling the refractive index (solvent) of the gain medium in microfluidic devices³⁶; multimodal lasing was achieved by the fabrication of superlattice cavities³⁷; and continuous wave lasing was realized using upconverting nanoparticles as gain medium³⁸. Also, fundamental work has established the robustness of lattice lasers against defects³⁹, has characterized their first-order coherence properties⁴⁰ and has shown that a single chain of nanoparticles could provide enough feedback to support lasing⁴¹. These results motivated the use of plasmonic lattices as an effective platform for light–matter coupling and ultralow-threshold lasing based on exciton–polaritons

a Lattice nanolaser structure



b Milestones of plasmonic lattice nanolasers

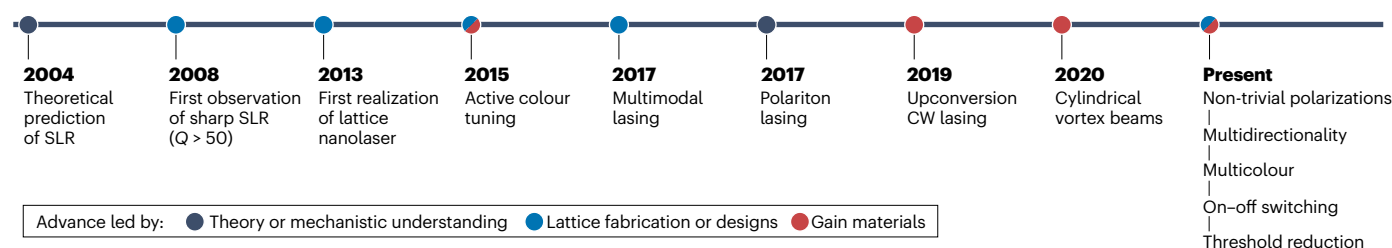


Fig. 1 | Overview of plasmonic lattice lasers and timeline of key advances.

a, Constituent components of plasmonic lattice lasers under optical excitation: nanoparticle (NP) lattices and gain materials. Plasmonic lattices consisting of noble or non-noble metal nanoparticles can be patterned on optically transparent substrates in both Bravais and non-Bravais geometries. Their

open-cavity architecture enables integration with diverse gain materials including organic dye solutions, solid-state, inorganic quantum-confined materials, and upconverting nanoparticles. **b**, Milestones of research on plasmonic lattice lasers. CW, continuous wave; Q , quality factor; SLR, surface lattice resonance.

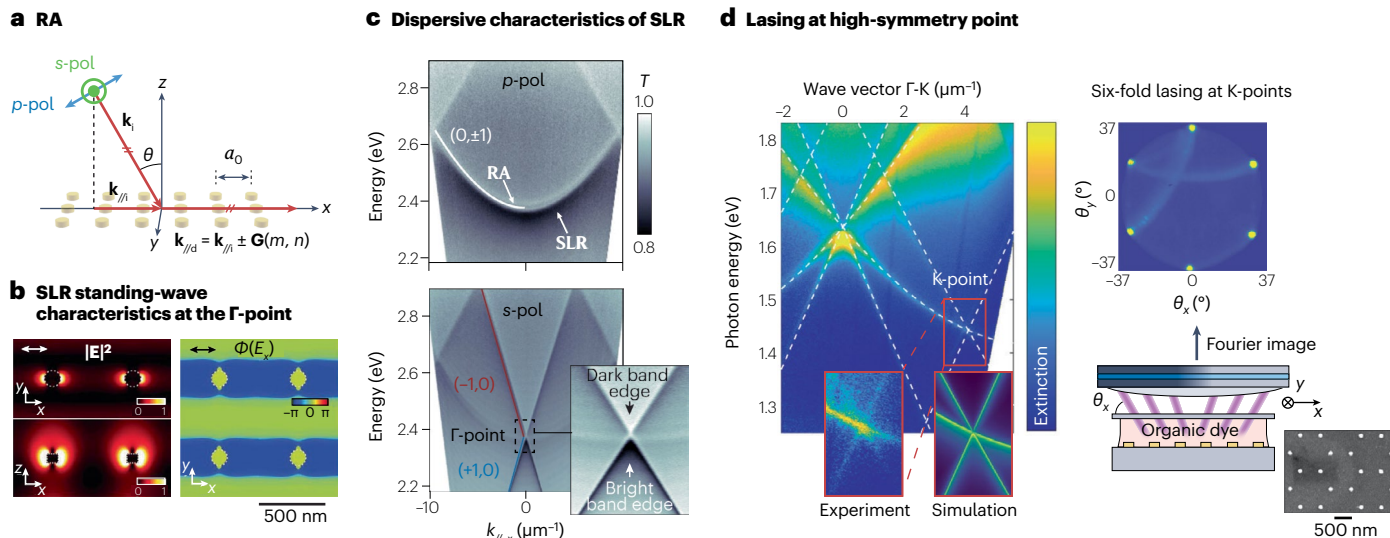


Fig. 2 | Plasmonic lattice laser fundamentals. **a**, Rayleigh anomaly (RA) excitation in a square nanoparticle lattice with period a_0 . Incident light arrives at an angle θ with p -polarization (p -pol) or s -polarization (s -pol) and wavevector $\mathbf{k}_i$ with in-plane wavevector ($k_{\parallel}$) along the x axis. At the diffraction cut-off condition, the incident light is diffracted as a RA mode, a surface-propagating wave. Such a condition can be found by solving $\mathbf{k}_{\parallel d} = \mathbf{k}_{\parallel i} \pm \mathbf{G}$, where $\mathbf{k}_{\parallel d}$ is the momentum of diffracted light and $\mathbf{G} = m\frac{2\pi}{a_x}\hat{x} + n\frac{2\pi}{a_y}\hat{y}$ is the reciprocal vector of the given lattice; a_x and a_y (both a_0 in this case) are the lattice periods along x axis and y axis, respectively; and (m, n) are integers denoting the diffraction orders. **b**, Calculated electric near-field intensity ($|\mathbf{E}|^2$) and phase ($\Phi(E_x)$) profiles of a square Au nanoparticle lattice ($a_0 = 600$ nm) in a refractive index $n = 1.52$ at the

surface lattice resonance (SLR) wavelength. **c**, Momentum-resolved transmission (T) spectra of a square Al nanoparticle lattice ($a_0 = 330$ nm) in $n \approx 1.6$ with SLRs under s -pol and p -pol. Calculated RAs (solid lines) exhibit either quadratic or linear dispersion bands. The inset highlights dark and bright band edges near the Γ -point. **d**, Momentum-resolved extinction of a plasmonic honeycomb lattice (left) and Fourier image with six-fold lasing spots at the K-points (right). The extinction zoom-in images highlight K-point band edges. The scanning electron microscopy image shows the honeycomb lattice. Panel **b** reprinted with permission from ref. 18, Elsevier, and reprinted from ref. 28, Springer Nature Limited. Panel **c** adapted with permission from ref. 42, American Chemical Society. Panel **d** adapted with permission from ref. 56, American Physical Society.

instead of photons^{42,43}. Moreover, plasmonic lattices have facilitated other collective phenomena beyond lasing, such as long-range energy transfer^{44,45} and Bose–Einstein condensation^{46,47}.

This Review discusses advances in plasmonic lattice lasers over the past 5 years (Fig. 1b). First, we summarize fundamentals and how unconventional lattice cavities and precise tuning of the unit cells can lead to additional degrees of control over lasing action. Second, we present design rules to realize specific lasing characteristics, including multicolour and multidirectional emission with non-trivial polarization and intensity distributions. Last, we discuss some open questions and prospects of the field.

Plasmonic lattice laser components

Plasmonic nanoparticle lattices

Light can be coupled into a lattice at the diffraction cut-off condition to form Rayleigh anomaly (RA) modes (Fig. 2a). The energy–momentum dispersion of these optical modes can be calculated by the empty lattice approximation, which only considers momentum conservation while neglecting contributions from nanoparticle light scattering and material properties^{48,49}. When the unit cells are made from metals, the nanoparticles can support localized surface plasmons whose properties depend on the nanoparticle size and shape⁵⁰. Their electromagnetic near-field response can be approximated to first order as in-plane electric dipoles; higher-order contributions become relevant as nanoparticle size increases^{50–52}. When there is spectral overlap between the localized surface plasmon (matter) and a RA mode (light), they can couple to form polaritons known as surface lattice

resonances (SLRs)^{49,53}. Overlap of the RA with the higher-energy tail of the localized plasmon results in an (upper) polariton with a narrow linewidth and stronger photonic character. The other (lower) polariton shows stronger plasmonic character and line shape and lower quality factor (Q) similar to that of the localized surface plasmon^{49,53}. At resonance, the SLR modes exhibit spatially delocalized, in-plane standing waves and sub-diffraction light confinement near the nanoparticles because of the RA and localized surface plasmon character, respectively (Fig. 2b).

Figure 2c shows how these features appear in experiment for a square Al nanoparticle lattice with periodicity $a_0 = 330$ nm under linearly polarized light. In energy-resolved and momentum-resolved transmission spectra, the SLRs closely follow the diffraction modes (solid lines), which are linear $(\pm 1, 0)$ under s -polarization and quadratic $(0, \pm 1)$ under p -polarization, except near high-symmetry points where band gaps form^{49,54}. The inset in Fig. 2c highlights the higher energy dark band edge and lower energy bright band edge at the Γ -point ($\mathbf{k}_\parallel = 0$). At band edges, the local density of optical states is maximized, which increases the stimulated emission rate when gain is present^{28,55}. Moreover, because band gaps typically occur at high-symmetry points, the far-field emission is determined by lattice symmetry, and multiple laser beams can be generated spatially and simultaneously depending on degeneracy. For example, honeycomb lattice cavities with C_3 rotational symmetry combined with organic dyes as gain medium can generate six laser beams from six K-points⁵⁶ (Fig. 2d). Hexagonal lattices with C_6 rotational symmetry have also produced six lasing beams but from six M-points³⁴.

Table 1 | Summary of gain material characteristics

Gain material	Size	Emission type	Material state	Unique lasing properties
Organic dyes	<2 nm	Molecular dipole transitions	Liquid, polymer-hosted film	Real-time wavelength tunability ³⁶ , polariton lasing ⁴³ , white light lasing ⁵⁸
Quantum dots	2–10 nm	Excitonic	Solid	Polarization tunability ⁶⁴ , polariton lasing ⁴²
Semiconductor nanocrystals	5–20 nm	Excitonic	Liquid, solid	Polarization tunability ⁶⁴ , chemically sensitive lasing ⁶⁸
Upconverting nanoparticles	5–100 nm	Atomic <i>f</i> -orbital transitions	Solid	Continuous wave lasing ³⁸ , lasing stability ³⁸

Gain materials

VCSEL and photonic crystal lasers typically use high-index semiconductors as gain materials⁵. By contrast, plasmonic lattice lasers have integrated discrete molecular or nanostructured emitters that can be concentrated within the localized electromagnetic hotspots near the nanoparticles. Table 1 compares common emitters used as gain media in plasmonic lasers along with their key properties enabling lasing characteristics.

Organic dyes. Organic dye molecules with high photoluminescence quantum yield (PLQY) were the first and smallest gain material tested in plasmonic lattice lasers²⁸. Their solubility in various organic solvents enables their processing into solid-state polymer films or solubilization as a liquid layer with dye concentrations ranging between 1 and 100 mM. However, their absorption and emission bands tend to be broad (>80 nm), and photostability is relatively low. When dyes are in solution, the gain medium can be introduced to the cavity through microfluidic channels, which results in robust lasing action because fresh molecules are available to interact with the nanoparticle lattice. In this architecture, the lasing wavelength can be tailored in real time by flowing different solvents with different refractive indices with the same dye molecules³⁶. Additionally, because the transition dipole moments of organic dyes depend on their molecular structure, the pump polarization determines which molecules are excited. Hence, specific optical modes associated with SLRs having different near-field or polarization profiles can be simultaneously coupled to the dye solution⁵⁷. Moreover, dye molecules are beneficial as gain material because of their miscibility with organic solvents and polymers. Multicolour emission and white-light lasing have been generated by solutions with different dyes in different relative concentrations⁵⁸, and polariton lasing has been facilitated by highly concentrated (>25 wt%) dye-doped films⁴³.

Quantum-confined emitters. When the size of an inorganic excitonic material is reduced to a few nanometres, the electron and hole wavefunctions become spatially localized within the emitter, leading to quantum-confinement effects^{59–61}. For example, CdSe quantum dots (QDs) and nanoplatelets (NPLs) exhibit size-dependent and narrow (<20 nm) absorption and emission bands spanning deep-ultraviolet to green wavelengths^{62,63}. Similar to molecular dyes, core-only QD and NPLs are prone to photodegradation, although shell layers (1–3 nm) can increase stability and PLQY while preserving quantum confinement⁶². In thin-film form (hundreds of nanometres), the high refractive index ($n \approx 1.8$ –2) of the emitters can support waveguide modes that can hybridize with SLRs to support lasing with cylindrical vector polarizations^{64,65}. Furthermore, the large oscillator strengths have been beneficial for polariton lasing⁴².

Nanocrystals and semiconductor emitters. When QDs are larger than the exciton Bohr radius, they are classified more broadly as

semiconductor nanocrystals⁶⁶. The absorption band broadens and approaches that of the bulk⁶⁰, and exciton radiative lifetimes, which are limited by multi-carrier Auger recombination, increase from ~10 ps to about 1 ns (ref. 63). Nanocrystals made from all-inorganic halide perovskites range in size between 5 and 20 nm and exhibit optical properties tuned primarily by composition instead of by quantum-size effects. Through stoichiometric control of either halide or cation species, perovskite emission across the visible spectrum is possible⁶⁷. Recently, these materials have enabled access to underexplored types of cavity modes⁶⁸ because of their high gain coefficients of over 450 cm^{-1} , with CsPbBr₃ nanocrystals exhibiting high PLQY, often exceeding 80%^{67,69,70}. In addition, core-shell nanocrystals composed of CdZnS–ZnS or CdSe–CdS can support gain coefficients higher than those of their core-only counterparts owing to the engineered epitaxy of the shell layer, which results in suppressed Auger recombination and chemical stability^{71–73}. These nanomaterials have even shown lasing action in colloidal solutions with plasmonic nanoparticle lattices⁷², with emitter concentrations markedly lower than those needed for molecules.

Upconverting nanoparticles. Upconverting nanoparticles (UCNPs) can convert low-energy photons into higher-energy emission but exhibit low PLQYs (0.1%–10%) that depend on size and morphology. The host lattice and dopant ion concentration of UCNPs determines upconversion wavelength and efficiency^{74,75}. Rare-earth-doped nanoparticles such as NaGdF₄ doped with Er³⁺ and Yb³⁺ have sizes between 5 and 100 nm, and smaller particles (<20 nm) often exhibit higher PLQY in the α -phase than in the β -phase from surface quenching effects^{74,75}. Because of their high photostability and long-lived 4f-orbital excited-state transitions, UCNPs can show robust emission under sustained excitation^{38,74}. This gain nanomaterial is the only type that has enabled continuous wave operation of a plasmonic lattice laser³⁸, wherein second-order temporal coherence was also confirmed.

Recent progress in plasmonic lattice lasers

Over the past five years, advances in the lasing characteristics of plasmonic lattice lasers have primarily relied on cavity architecture. We highlight three structural features: unit cell design for beam engineering, lattice symmetry and combinations for chromatic and directional emission, and nanoparticle materials for threshold engineering. Plasmonic lattice lasers can exhibit tunable lasing angles, emission from ultraviolet to near-infrared wavelengths, and non-trivial polarization states – from the same device – which are characteristics not accessible to VCSELs.

Beam polarization engineering

Far-field lasing beam characteristics, such as polarization and intensity profile, are determined by the cavity mode. Because transverse magnetic (TM) or transverse electric (TE) waveguide modes in Fabry–Perot resonators are only linearly polarized, lasing outputs from VCSELs are also linearly polarized. Therefore, to achieve additional polarization

control, external components including polarizers or spatial light modulators are required⁷⁶. By contrast, plasmonic lattice lasers offer multiple ways to tune lasing properties via the electromagnetic near-field properties of nanoparticle unit cells, lattice symmetry and waveguide-hybridized SLR modes.

The simplest plasmonic lattice cavities for lasing applications are those embedded in a uniform refractive index. However, when materials with high refractive index ($n > 1.7$) are used as gain medium, their films can support waveguide modes that depend on thickness. Figure 3a shows how waveguide modes from QD thin films (red solid line)⁶⁴ can hybridize with SLRs to form waveguide-SLR (W-SLR) cavity modes^{64,65,77}. Compared to index-matched SLR modes (Fig. 2a), W-SLRs support electric near-field distributions that extend throughout the high-index layer, whereas the near-field hotspots remain near the nanoparticles. Because thin-film TM and TE waveguide modes generally exhibit quadratic behaviour⁷⁸, W-SLR bands have complex quadratic and linear behaviour in the photonic dispersion diagrams; band gaps still form at high-symmetry points and at band crossings. Although TE and TM polarizations are orthogonal, linear W_{TE} -SLRs (W_{TM} -SLRs) and quadratic W_{TM} -SLRs (W_{TE} -SLRs) can be observed simultaneously under s -polarized (p -polarized) excitation (Fig. 3b). The waveguide component can further decrease radiative losses from SLRs leading to additional higher- Q band edges^{64,65,77,79}. Al nanoparticle square lattices combined with CdSe QD films exhibit lasing with cylindrical vector beam profiles facilitated by the near-field coupling supported

by W-SLRs⁶⁴ (Fig. 3c). Angle-resolved transmission measurements show that lasing (Fig. 3c, red solid curve) occurs at the dark band edge. Because the dark W_{TE} -SLR band edge can only couple to free-space radiation at angles other than 0° ($k_{\parallel} = 0$), lasing emission is only observed at $k_{\parallel} \neq 0$, leading to a donut-shaped intensity distribution. Notably, these beams exhibit unique polarization states beneficial for single-particle trapping and super-resolution imaging, which traditionally require fibre optics or bulky optical set-ups to achieve^{80,81}. As the QD film thickness increases, the cavity mode that supports lasing changes from W_{TE} -SLR to W_{TM} -SLR, which changes the polarization state from azimuthal to radial. These results demonstrate the profound impact that the near-field properties of the cavity modes have on the far-field lasing characteristics.

Expanding on this concept, plasmonic lattices combined with high-refractive-index perovskite nanocrystal films were used to generate cylindrical vortex laser beams with hybrid electric (HE) polarization⁸². HE-polarized beams are indexed by their polygonal polarization about the beam centre^{83,84}. The top panel of Fig. 4a displays an HE beam profile from a square lattice with C_4 symmetry under a linear polarizer that transforms the donut-shaped intensity into a two-lobed distribution. When the same lattice and perovskite gain sample is rotated by 45° , a different HE beam square-symmetric polarization intensity distribution arises (Fig. 4a, middle). The origin of this pattern can be attributed to W-SLR modes with HE polarization because when nanoparticles increase in size and support in-plane quadrupolar localized plasmons, they can

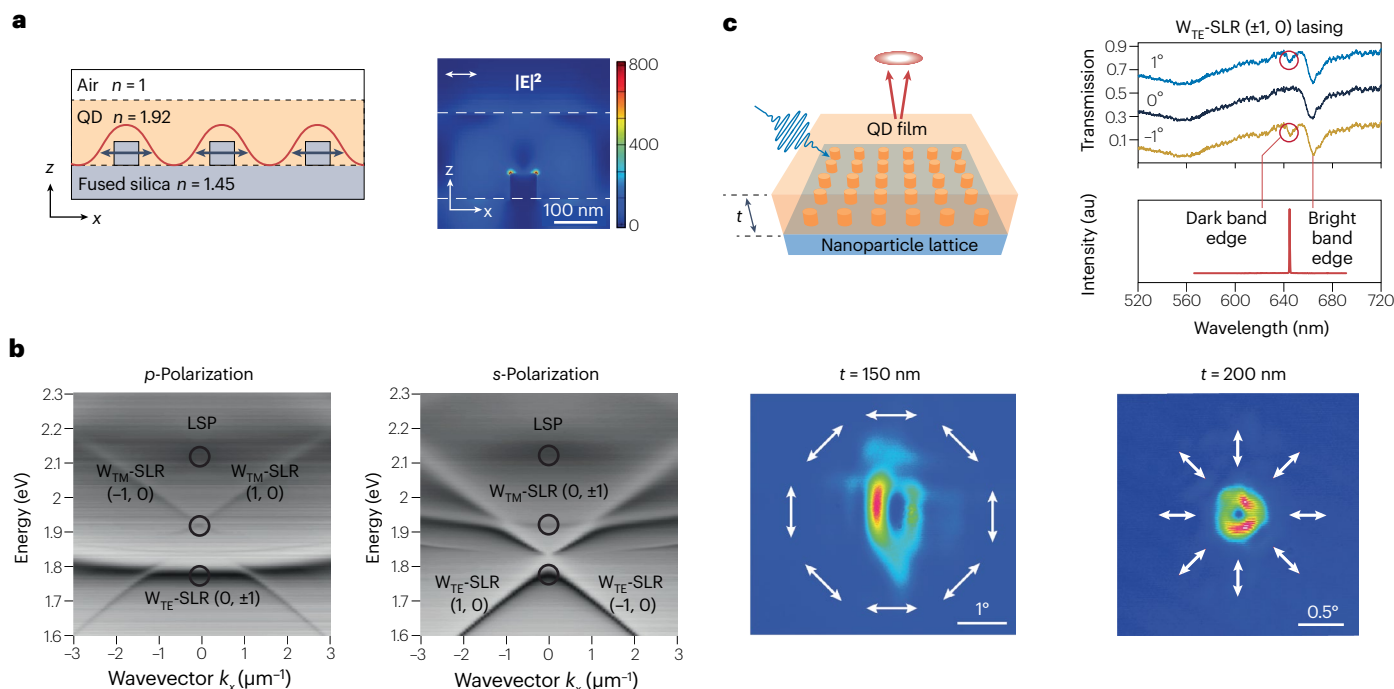
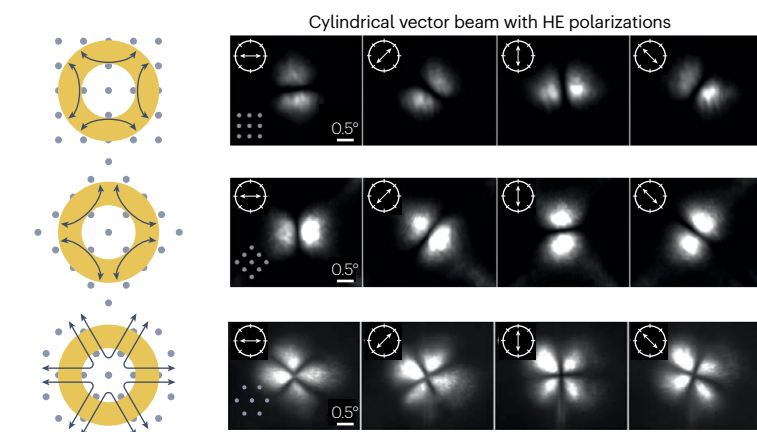


Fig. 3 | Optical modes in plasmonic lattice lasers with high-index gain layers. **a**, Waveguide and surface lattice resonance (SLR) coupling scheme with superstates with high refractive index n ; the solid line indicates the waveguide mode and the black arrows the in-plane dipole of nanoparticles (NPs) (left). Calculated electric near-field intensity at the waveguide-SLR (W-SLR) wavelength (right) shows that field enhancement is not limited to the NP vicinity but is found in the high-index layer owing to the waveguide response. **b**, Energy-resolved and momentum-resolved transmission spectra of a Ag NP square lattice coated with a CdSe quantum dot (QD) film under p -polarized and s -polarized illumination with

in-plane wavevector along x axis. The circles highlight the modes at $k_x = 0$. **c**, The generation of a donut-shaped lasing beam from W-SLRs (top left) and lasing and transmission response of W_{TE} -SLR ($\pm 1, 0$) under different incidence angles (top right). When the QD film thickness changes from $t = 150$ nm to $t = 200$ nm, the beam polarization changes from azimuthal to radial, as observed in angle-resolved photoluminescence measurements (bottom). White arrows denote the beam polarization distribution. LSP, localized surface plasmon. Adapted with permission from ref. 64, American Chemical Society.

a Polarization control: W-SLR and lattice symmetry



b More complex profiles from moiré lattices

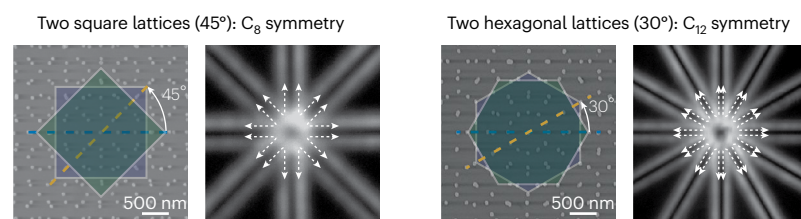
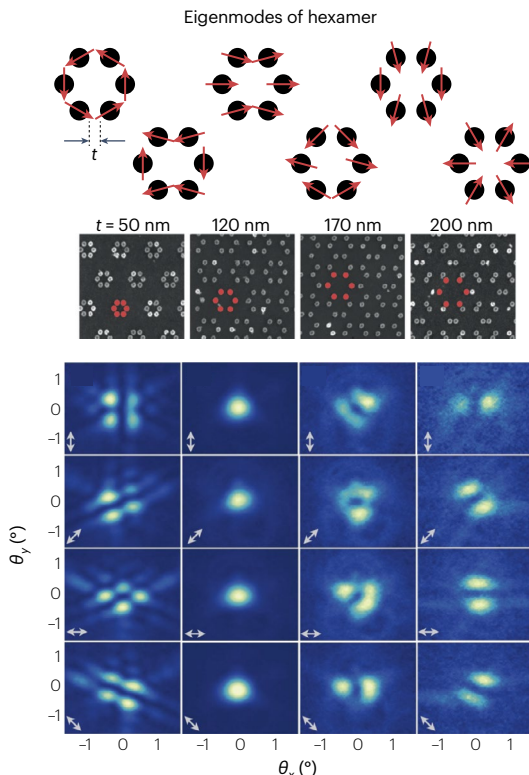


Fig. 4 | Non-trivial polarization state generation from lattice symmetry and unit cell design. **a**, Hybrid electric (HE)-polarized cylindrical vector beams supported by waveguide-surface lattice resonance (W-SLRs) from a square (top), a 45°-rotated square (middle), and an hexagonal lattice (bottom). The schemes illustrate the polygonal symmetry of the polarization state (black arrows) as indicated by the 2D angle-resolved images filtered with a linear polarizer whose axis is parallel to the white arrows. **b**, Scanning electron microscopy (SEM) images and 2D angle-resolved lasing beam images of C_8 and C_{12} symmetric moiré lattices made by superimposing two square lattices at a 45° angle and two hexagonal lattices at a 30° twist angle in the same plane; shaded regions denote

hybridize simultaneously to both TE and TM waveguide modes⁸². In addition, lattice rotational symmetry was found to be directly related to the polarization state. For example, the bottom panel of Fig. 4a shows how a higher-symmetry C_6 (hexagonal) lattice produces a four-lobed hexagonally symmetric polarization profile (black arrows). Increasing the rotational symmetry of the cavity even further through the design of moiré lattices made of two square and two hexagonal lattices at offset angles of 45° and 30°, respectively, to C_8 (octagonal) and C_{12} (dodecagonal) increased the HE polarization state (Fig. 4b). Hence, rotational symmetry of lattice cavities enables a scalable and versatile design principle for advanced polarization engineering.

Nanoparticle oligomers, which consist of multiple nanoparticles placed in proximity, have also been used to shape the polarization state of the lasing beam because these patterns can support multiple eigenmodes (localized surface plasmon modes)⁸⁵. The top panel of Fig. 4c depicts six different in-plane eigenmodes of a hexamer structure, wherein the red arrows denote the dipole orientation of each particle. Using dye solutions as gain medium and Au nanoparticle hexamers arranged in triangular lattices, lasing with non-trivial polarization

c Polarization control: oligomer unit cells



the geometry of the base lattices. Double-headed white arrows indicate the polarization distribution. **c**, Hexamer in-plane eigenmode representation (top); red arrows represent in-plane dipole orientation of each nanoparticle and t is the gap distance. SEM images and 2D angle-resolved beam intensities of hexamer lattices with different t values (middle). Unit cell structures are highlighted with red dots for clarity. The intensity images were filtered by a linear polarizer whose axis is parallel to the white arrows (bottom). Panels **a** and **b** adapted with permission from ref. 82, American Chemical Society. Panel **c** adapted with permission from ref. 86, American Physical Society.

states was demonstrated⁸⁶. Two-dimensional angle-resolved images at different polarizations show that varying the gap size between 45 and 200 nm while maintaining the same lattice periodicity results in different polarization outputs (Fig. 4c, middle and bottom). Because unit cell modifications change the eigenmode Q , the SLR mode with the highest Q will support lasing as a result of mode competition. These results demonstrate an alternative route to polarization engineering using the numerous eigenmodes of plasmonic oligomers.

Multidirectional and multichromatic emission engineering

Early demonstrations of plasmonic lattice lasers that could produce multiple emission beams of different colours relied on two different designs for the cavity: superlattices or non-Bravais lattices. For superlattices, in-plane Bragg scattering between micrometre-sized patches of nanoparticle lattices patterned with microscale periodicity formed superlattice band edges about a dominant SLR mode from the nanoparticle lattice⁸⁷. These superlattice SLRs combined with organic dyes as gain medium could support multimode lasing but had limited spectral and angular range (around ± 20 nm, $\pm 5^\circ$)³⁷. For non-Bravais lattices, quasi-periodic

and aperiodic lattices with dense optical band structures have shown multi-beam lasing³⁵, and aperiodic superlattices have resulted in multichromatic and directional lasing⁸⁸. Although structure factor analysis was used to identify the modes that support lasing in these non-Bravais structures^{35,88}, a model that can predict which of the multiple band edges contributes most to the lasing emission would be beneficial.

Recently, the use of moiré lattices as optical cavities demonstrated that lasing is favoured for modes sharing a rotational symmetry relationship. An eight-fold rotationally symmetric moiré lattice can be fabricated through two patterns of a square base lattice at a specified (45°) twist angle (Fig. 5a). Combined with different dye solutions, moiré lattices exhibit lasing over distinct spectral ranges and at multiple emission angles determined by the lattice symmetry⁸⁹. For example, angle-resolved and wavelength-resolved photoluminescence images indicate lasing emission at -1.7 eV and at five different angles (0°, ±38°, ±57°). Empty lattice calculations were used to identify the lasing modes and other isoenergetic modes and revealed that lasing was only found from collections of modes in reciprocal space that formed congruent octagons (eight-fold symmetric). These results suggest that lattice rotational symmetry can be used as a design rule to favour lasing at specific band edges.

Combining the superlattice cavity concept with different nanoparticle materials and different lattice periodicities provides

a straightforward approach to engineer lasing from specific SLR modes. Figure 5b shows a schematic of a bimetallic cavity structure consisting of interwoven Al and Au nanoparticle lattices with periods $a_{\text{Al}} = 400$ nm and $a_{\text{Au}} = 500$ nm, each supporting ($\pm 1, 0$) SLRs at $\lambda_{\text{Al}} = 583$ nm and $\lambda_{\text{Au}} = 718$ nm. Different metal nanoparticle lattices patterned into quasi-random areas can support single band-edge modes at the Γ -point that can overlap with different dyes⁹⁰. Because the optical constants of Al and Au are suitable for supporting strong plasmonic responses at visible and near-infrared wavelengths, respectively, single substrates composed of multiple lattices can support multiple, sharp SLRs⁹¹. When combined with solutions of two different dyes (C540 and LD700), a bimetallic lattice can support lasing at visible and near-infrared wavelengths both independently and simultaneously. Therefore, the lasing mode energy and angle can be accurately designed. Because plasmonic lattice lasers can operate even after removing 90% of the particles in a lattice cavity, this multimetallic approach could be extended to up to ten distinct regions with different combinations of metals.

Besides combining different nanoparticle lattices in the same plane, nanoparticle lattices can also be stacked vertically. Figure 5c depicts this approach, wherein solutions of dye molecules (DCM, C500 and C480) were sandwiched between square lattices of Al nanoparticles with $a_0 = 350$ nm and 450 nm. The 350-nm lattice ($\pm 1, 0$) band edge was

a Multidirectional lasing: moiré lattice

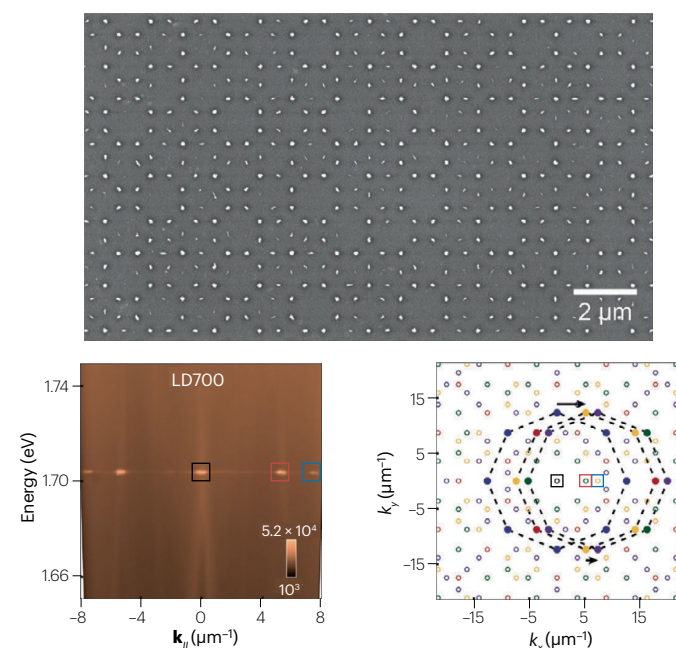
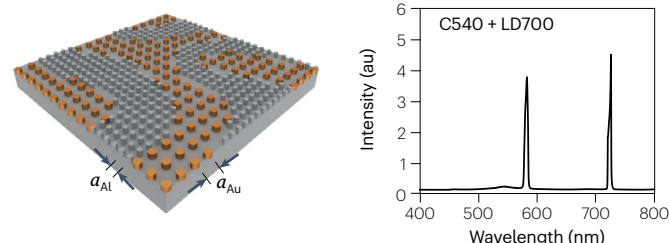


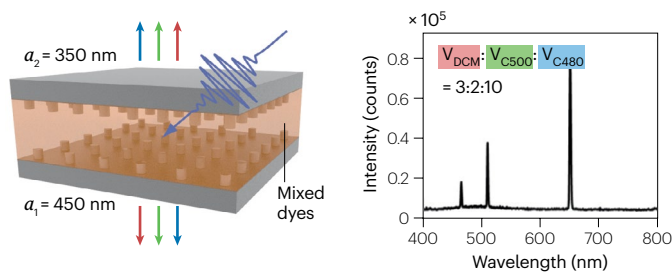
Fig. 5 | Multidirectional and multichromatic output generation.

a, Multidirectional lasing from a moiré lattice. A scanning electron microscope (SEM) image of an Al nanoparticle moiré lattice with eight-fold rotational symmetry (top). When the lattice is combined with a LD700 dye solution and under pulsed excitation, five lasing spots are recorded in the energy-resolved and momentum-resolved photoluminescence spectra (bottom left). The directionality of the lasing spots can be predicted from the points in the reciprocal space forming congruent octagons, centred around the spots marked with coloured squares (bottom right). **b**, Multichromatic lasing of quasi-random multimetallic lattices made of Au and Ag nanoparticles with periods $a_{\text{Al}} = 400$ nm

b Multichromatic lasing: multimetallic lattice



c Multichromatic lasing: stacked lattices



and $a_{\text{Au}} = 500$ nm, respectively. The lasing spectrum shows Γ -point, two-colour, simultaneous lasing when the bimetallic substrate is combined with a C540 and LD700 dye solution. **c**, Multichromatic lasing of two stacked Al nanoparticle lattices with period a_1 and a_2 combined with a liquid DCM, C500 and C480 dye mixture (3:2:10 ratio). Under a 400-nm pulsed excitation, the lasing spectrum shows three simultaneous lasing beams emitted normal to the sample. Panel **a** reprinted with permission from ref. 89, Wiley. Panel **b** reprinted with permission from ref. 90, American Chemical Society. Panel **c** adapted with permission from ref. 58, Wiley.

selected to overlap with the photoluminescence of C500 and to support single-mode lasing. The 450-nm lattice can show single-wavelength emission, either from the photoluminescence of C480 overlapping with the $(\pm 1, \pm 1)$ mode or from that of DCM overlapping with the $(\pm 1, 0)$ mode, or dual-wavelength lasing from both dyes simultaneously when they are in the same solution. Dye mixing ratios can be empirically adjusted to generate a combined lasing emission within the Commission Internationale de l'Éclairage (CIE) colour space. When the nanoparticle lattices were stacked and three different dyes were in the same solution, blue, green and red lasing was possible; the far-field intensity profile showed how the overlapped emission could sum up to white light⁵⁸. However, such dye mixtures have drawbacks because their broad emission and absorption bands can result in quenching because of reabsorption and fluorescence resonance energy transfer^{92–94}.

Threshold engineering

Nanoparticle materials. Given that the rate of stimulated emission scales with cavity mode quality, most plasmonic lattice lasers have been designed with nanoparticles made of noble metals. However, non-noble metals can also realize lasing action despite their higher ohmic losses. One example is Cu nanoparticles, whose Drude parameters are very similar to those of Au (ref. 91), but which has been underexplored because surface oxidation degrades its plasmonic properties²⁹. Using Cu nanoparticles as catalysts for selective and conformal growth of few-layer graphene (~3 nm thick), which also inhibited oxidation⁹⁵, reduced the lasing threshold⁹⁶. Graphene-coated Cu nanoparticle lattices show a more efficient lasing response than either bare Cu nanoparticles or Cu nanoparticles coated with a dielectric layer of the same thickness as graphene (Fig. 6a). Strikingly, graphene not only mitigates dye–metal quenching but also increases the local dye concentration at nanoparticle surfaces via π – π interactions, with a ~60% reduction in lasing threshold compared to bare nanoparticles.

Another instance of non-noble metals for lattice cavities is Co/Pt multilayer nanoparticles (Fig. 6b), wherein external magnetic field control over lasing with dye molecules as gain medium was demonstrated⁹⁷. In this system, a circularly polarized pump not only excited the dyes but also induced circular motion of free electrons, generating a magnetic field around the nanoparticles. The lasing characteristics were modulated by interactions between the external field-driven magnetization and the pump-induced magnetic field. The ‘down’ magnetization state required less energy for lasing with the left circularly polarized pump and vice versa for the opposite handedness. Under a net magnetic field and circularly polarized excitation, eigenmode degeneracy in nanoparticles was lifted, leading to distinct responses such as magnetic circular dichroism⁹⁸ that can affect the lasing response. Modulating lasing thresholds by controlling either surface chemistry or ferromagnetic properties of the nanoparticle lattice cavity highlights how materials engineering can open additional lasing capabilities. Looking ahead, functionalizing nanoparticle surfaces with responsive ligands and catalytic shells^{99,100} may enable effective sensing and photocatalysis driven by lasing action.

Polariton lasing. Another approach to reduce lasing thresholds for plasmonic lattice cavity architectures is to use polaritons instead of photons. Exciton–polaritons, quasiparticles resulting from the strong coupling between excitons and photons, provide opportunities to lower thresholds because polariton lasing does not require population inversion¹⁰¹. The polariton coupling strength between

photons in the cavity and excitons of the emitters is determined by the Rabi splitting

$$\Omega = \mu \cdot \sqrt{\frac{2\hbar\omega}{\epsilon_0 V_c}} \cdot \sqrt{N}$$

where μ is the transition dipole moment, $\hbar\omega$ the cavity resonance, V_c the mode volume and N the number of emitters coupled to the cavity¹⁰². When Ω exceeds the average losses of the uncoupled emitter and cavity mode, the system enters the strong-coupling regime¹⁰³ and forms two new eigenstates, a lower polariton and an upper polariton, that can be observed in angle-resolved transmission measurements. When a critical polariton population accumulates (usually in the lower polariton), their spontaneous radiative decay results in coherent light emission or polariton lasing^{42,43,104,105}. The polariton population can be increased by two different mechanisms: vibration-assisted relaxation and polariton–polariton scattering, depending on the type of gain material.

Organic dye emitters favour vibration-assisted relaxation (Fig. 7a), wherein uncoupled excitons at the vibrational ground state ($v = 0$) can be directly transferred to the lower polariton via the first vibrational excited state and then relaxed further. This process is most effective when the energy difference between the cavity mode and dye absorption band (Δ_1) matches that between adjacent vibronic sidebands (Δ_2). Figure 7b illustrates this mechanism for a rylene dye whose emission increases non-linearly as a function of excitation fluence wherein this condition is met⁴³; this selection rule ($\Delta_1 = \Delta_2$) has been shown to govern polariton lasing in organic systems¹⁰⁵. By contrast, the large Bohr radii of Wannier–Mott excitons in semiconductor emitters favours polariton–polariton scattering for efficient polariton accumulation^{101,106} (Fig. 7c). In this mechanism, polaritons reduce their energy via polariton–polariton scattering until reaching a local minimum. This energy relaxation pathway was recently confirmed in core-only CdSe NPLs combined with Al nanoparticle lattices. Room-temperature polariton lasing with thresholds as low as ~6 $\mu\text{J cm}^{-2}$ was demonstrated, which is around 25 times lower than that observed in similar CdSe NPL plasmon lattice lasers operating in the photon lasing regime^{42,107}. Notably, the polariton lasing wavelength showed gradual blueshift and peak broadening as the pump fluence increased (Fig. 7d), indicating coulombic interactions in densely populated polariton states¹⁰⁸.

Future perspectives and challenges

Open questions around lattice nanolasing mechanisms

Progress in plasmonic lattice lasers has been accompanied by several puzzles. Plasmonic lattices currently rely on lasing models that make assumptions about single unit cells and that cannot fully capture experimental finite lattice effects¹⁰⁹. Also, whereas most work on plasmonic lattice lasers has focused on optical feedback from SLRs, recent work suggests that feedback from RA modes or other modes can also lead to lasing action^{42,72,110}. Contrary to expectations that optical gain is driven by intense and highly localized fields near the plasmonic nanoparticles, lasing has been demonstrated in systems wherein the dye gain solution was situated at a distance several orders of magnitude larger than a_0 (~170 μm) away from the lattice cavity¹¹⁰. Simulations indicate that the out-of-plane interactions could be mediated by dark SLRs that show long spatial and temporal coherence¹¹⁰.

Additionally, lasing from plasmonic lattices has been demonstrated at crossings of RA modes instead of at SLR band edges⁷².

a Threshold reduction through surface chemistry

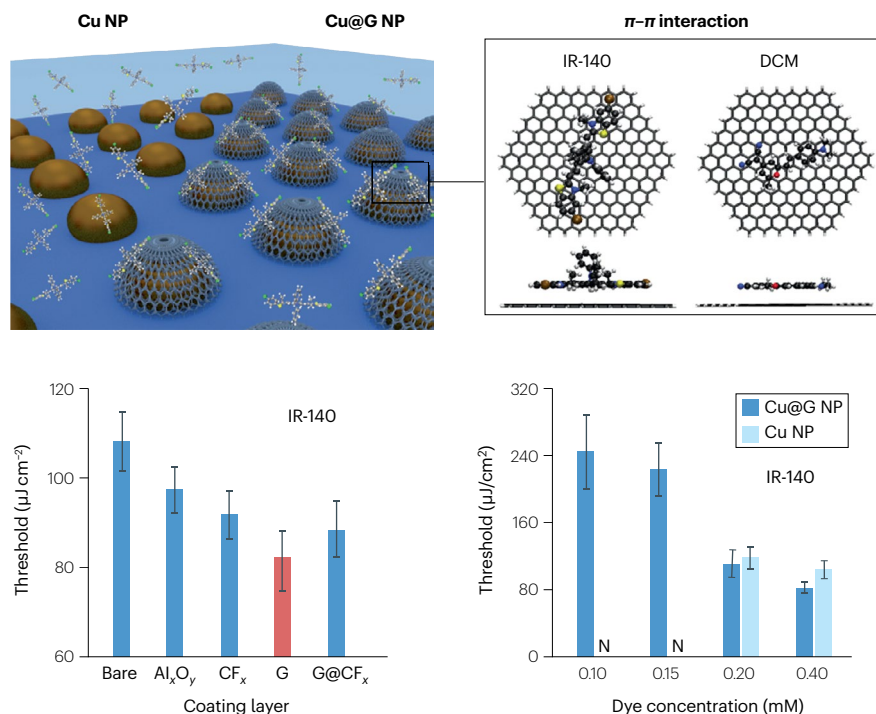
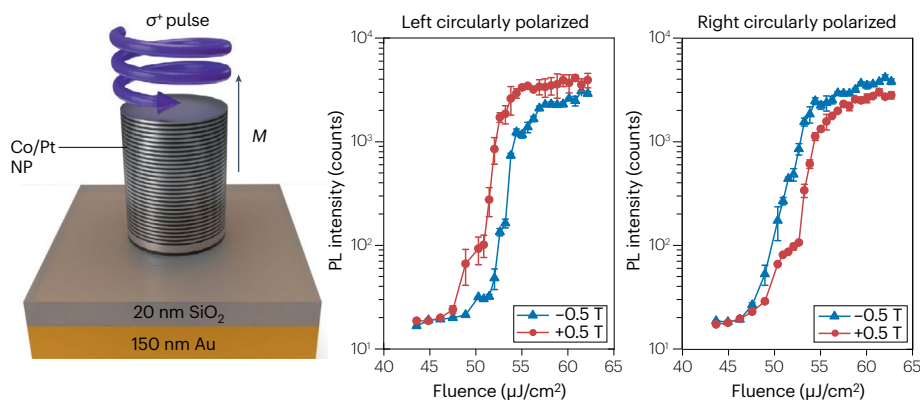


Fig. 6 | Laser engineering via functional NPs.

a, Nanoparticle (NP) lattice lasers made of Cu NPs covered in graphene (top left) exhibit lower thresholds than plasmonic lattice lasers made of bare Cu NPs and of Cu NPs covered in Al_2O_3 , CF_x and $\text{G}@CF_x$ (bottom left). Molecular dynamics simulations predict improved aromatic dye adsorption on graphene via π - π interactions (top right), which could account for operation at lower dye concentrations (bottom right) and the threshold reduction. N in the threshold histogram means absence of lasing action. **b**, Ferromagnetic Co-Pt multilayer NP lattice lasers under circularly polarized excitation display two different lasing thresholds depending on the magnetization state (M) of NPs, as seen in the photoluminescence (PL) response as a function of the excitation fluence for $\pm M$ states and for left and right circularly polarized pulses. Panel **a** adapted with permission from ref. 95, PNAS, and adapted from ref. 96, CC BY-NC-ND 4.0. Panel **b** adapted from ref. 97, Springer Nature Limited.

b On-off modulation through magnetic response of NPs



Plasmonic lattices integrated with CdZnS–ZnS nanocrystal solutions exhibited lasing at the degenerate RA modes despite the intense spontaneous emission from SLR bands at longer wavelengths. Photoluminescence measurements revealed that the RA bands had narrower linewidths, which suggests higher Q values. These experimental results are supported by coupled-dipole calculations indicating that RAs can coexist with SLRs^{53,72}. These results indicate that the optical response of plasmonic lattices cannot be fully explained using a Jaynes–Cummings model¹¹¹ wherein two uncoupled modes – RA and localized surface plasmon – hybridize to form two strongly coupled (SLR) states.

Improved mechanistic understanding is critical to explore unconventional lattice designs that may lead to unanticipated lasing responses. Recently, plasmonic quasicrystals with only rotational

symmetry have shown lasing action despite the seemingly limited optical feedback and emission directionality¹¹². Controlling the rotational symmetry of quasicrystals has resulted in lasing with very high topological charge numbers that exceed what is typical for plasmonic and dielectric moiré structures composed of periodic sub-lattices^{82,113}. Inspiration can also be drawn from designs of dielectric topological lasers for topological protection¹¹⁴. For example, plasmonic lattices offer the potential to realize unconventional responses such as non-Hermitian skin effects¹¹⁵ and excitation of edge states¹¹⁶. Au nanoparticle quadrumer lattices have been shown to mediate coupling not only between adjacent unit cells (that is, a tight-binding model) but also between unit cells farther away, indicating that edge states can be excited by pumping the entire lattice¹¹⁶.

Polariton accumulation: vibration-assisted relaxation

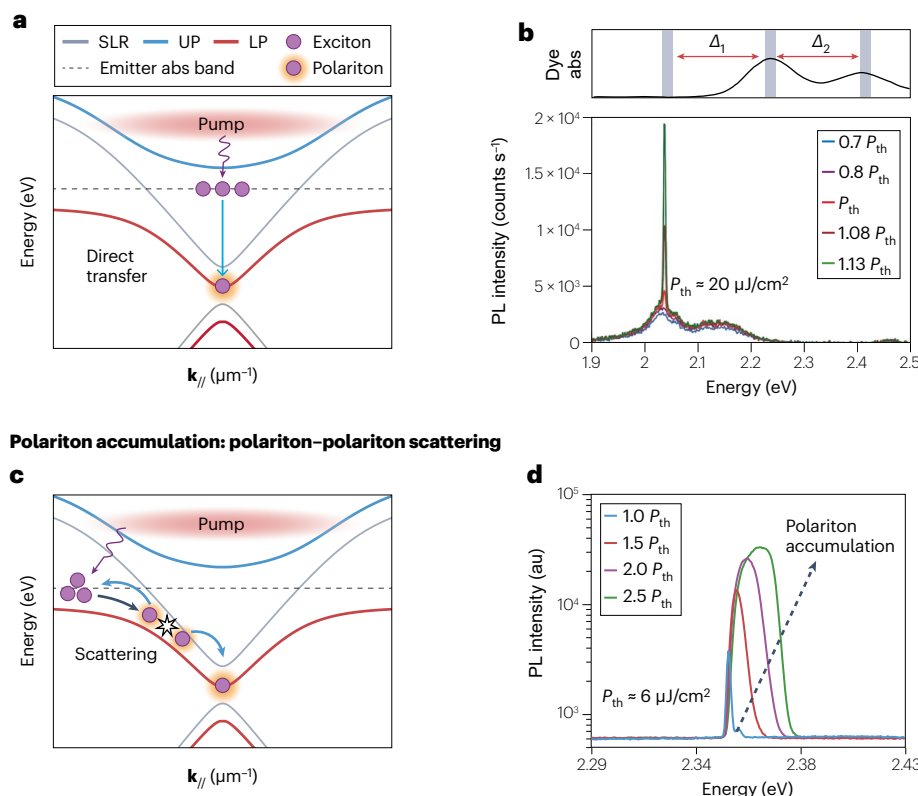


Fig. 7 | Polariton lasing from organic and inorganic emitters.

a, Lower polariton (LP) accumulation through vibrational-assisted relaxation for organic emitters. The pump generates excitons and polaritons that result in direct polariton accumulation at the LP minima via polariton–phonon interactions.

b, Photoluminescence (PL) of a polariton nanoparticle lattice laser consisting of a rylene-doped polymethylmethacrylate film coupled to a Ag nanorod lattice for excitation fluence values below and above threshold (P_{th}). The dye absorption spectrum showing vibronic sidebands is shown on top. The energy difference between the first vibronic sideband and the lasing peak (Δ_1) and that between the two adjacent vibronic sidebands (Δ_2) are highlighted.

c, LP accumulation through polariton–polariton scattering for inorganic emitters. The pump populates polariton bands that relax their energy along the band primarily through polariton–polariton scattering events until the local minima are reached.

d, Above-threshold polariton lasing spectra of CdSe nanoparticles coupled to an Al nanoparticle lattice. As the fluence increases, gradual emission blueshift and broadening are observed as expected from polariton–polariton coulombic interactions in densely populated polariton states. abs, absorbance; SLR, surface lattice resonance; UP, upper polariton. Panel **b** adapted with permission from ref. 43, © Optica Publishing Group. Panel **d** adapted with permission from ref. 42, American Chemical Society.

Need for more sophisticated lattice nanofabrication

Plasmonic lattices with unconventional architectures, such as those with oligomer unit cells^{117–120}, vertically stacked nanoparticle lattices^{121–124}, and unit cells with vertically offset nanoparticles^{125,126}, exhibit complex optical responses that are extremely sensitive to parameters such as nanoparticle shape and size, oligomer geometry, interparticle gap spacing, or vertical and lateral displacement. In particular, plasmonic nanoparticle oligomers are emerging as promising lattice building blocks (Fig. 4c). In a plasmonic lattice consisting of monomer unit cells, the nanoparticles are coupled through far-field scattering. For oligomers with interparticle gaps of less than tens of nanometres, the localized surface plasmons from individual particles couple in the near field¹²⁷, resulting in multipolar oligomer resonances with unique responses such as optical chirality¹¹⁷ and super-radiance and sub-radiance¹¹⁸. As the gap size decreases further (~ 10 nm)⁸⁵, the near-field coupling not only strongly confines the electric field within the gaps but also induces strong optical magnetism^{119,120}. Such optical responses depend sensitively on nanoparticle shape and size, oligomer geometry and gap size, thus requiring precise parameter control and high-fidelity patterning⁸⁶.

Conventional nanofabrication based either on only top–down (such as electron-beam lithography) or only bottom–up (such as colloidal self-assembly) approaches are not well-suited for the fabrication of wafer-scale oligomer lattices because of pattern distortions induced by substrate charging and the low uniformity of assembly, respectively. Alternative pattern generation and transfer techniques based on nano-imprint lithography have enabled large-area and uniform fabrication of plasmonic lattices through parallel processes^{128,129}. Also, combining

the advantages of top–down and bottom–up methods offers promise for the creation of oligomer lattice cavities. For example, advances in using chemical modifications, including metal shell growth and surface ligand functionalization combined with lithographically patterned nanoparticle seeds, have resulted in large-area lattices of nanoparticle oligomers organized in two and three dimensions^{130–132}.

Electrically driven plasmonic lattice lasers

The beam-profile tunability of plasmonic lattice lasers is advantageous for multi-polarization and wavelength LIDAR (light detection and ranging)¹³³, optical computing¹³⁴ and bio-imaging, on-chip devices¹³⁵. However, one critical operational requirement is to achieve electrically rather than optically driven excitation, which has not yet been demonstrated. The main challenges are the limited electroluminescence quantum efficiencies of typical gain materials (Table 1) and issues in achieving electrode–lattice integration for sufficient electron injection¹³⁶. Combining recent advances in emitters and LED electrical-injection architectures with plasmonic lattices could make electrically driven plasmonic lasers possible. Continuously graded colloidal QDs with reduced Auger recombination combined with photonic waveguides and Fabry–Perot cavities could be electrically pumped and exhibit amplified spontaneous emission¹³⁷. Also, when integrating such emitters with plasmonic lattice cavities that can support electron injection, an inverted LED architecture with 2D electronic materials or low-index indium tin oxide (ITO, $n \approx 1.6$) electrodes could be suitable^{138,139}. Recently, the demonstration of electrochemical tunability of strong coupling in CdSe NPLs and plasmonic Al lattices on ITO–glass substrates highlighted the benefits of integrating lattices

into electrode designs¹⁴⁰. For laser applications, a similar type of electrode would ensure that W-SLRs are only excited in the high-index gain layer, which would maximize gain and avoid optical losses from mode excitation in the adjacent current-confinement layers^{138,139}. Finally, integrating plasmonic lattices with gain materials used in commercial VCSELs (such as GaN, InGaAs and InP) offers a promising strategy to achieve electrically operated plasmonic lasers^{141,142}.

In summary, we have highlighted how plasmonic lattice lasers have emerged as a versatile platform for coherent light control and manipulation. Recent innovations in lattice cavity design, nanoparticle materials, and integration of diverse gain nanomaterials have significantly broadened the scope of desirable lasing characteristics, enabling non-trivial beam polarization, multidirectional emission, wavelength tunability and threshold control. Although challenges in the fabrication of complex lattices and in the development of electrically driven systems remain, progress in this area offers exciting opportunities for future optoelectronic devices and light sources offering unprecedented control and flexibility.

Published online: 13 May 2025

References

- Gordon, J. P., Zeiger, H. J. & Townes, C. H. The maser — new type of microwave amplifier, frequency standard, and spectrometer. *Phys. Rev.* **99**, 1264–1274 (1955).
- Campbell, C. J. Ruby laser prototype. *Columbia University* <https://www.vagelos.columbia.edu/departments-centers/ophthalmology/about-us/our-history/collections/ruby-laser-prototype> (1961).
- Schwartz, M. I., Reenstra, W. A., Mullins, J. H. & Cook, J. S. Atlanta fiber system experiment: the Chicago Lightwave Communications Project. *Bell Syst. Tech. J.* **57**, 1881–1888 (1978).
- Weightman, G. The history of the bar code. *Smithsonian Magazine* <https://www.smithsonianmag.com/innovation/history-bar-code-180956704/> (2015).
- Lotsch, H. K. V. et al. VCSELs Vol. 166 (Springer, 2013).
- Chen, Q. et al. Highly efficient vortex generation at the nanoscale. *Nat. Nanotechnol.* **19**, 1000–1006 (2024).
- Fang, X., Ren, H. & Gu, M. Orbital angular momentum holography for high-security encryption. *Nat. Photon.* **14**, 102–108 (2020).
- Liu, X., Li, A., Jiang, X., Yang, H. & Li, Y. Cascaded-prism multi-mode beam scanning method for three-dimensional imaging lidar. *Appl. Opt.* **63**, 5670 (2024).
- Schwarz, B. Mapping the world in 3D. *Nat. Photon.* **4**, 429–430 (2010).
- Guan, J. et al. Light-matter interactions in hybrid material metasurfaces. *Chem. Rev.* **122**, 15177–15203 (2022).
- Overvig, A. & Ali, A. Diffractive nonlocal metasurfaces. *Laser Photon. Rev.* **16**, 2100633 (2022).
- Koshelev, K. & Kivshar, Y. Dielectric resonant metaphotonics. *ACS Photon.* **8**, 102–112 (2021).
- Lončar, M., Yoshie, T., Scherer, A., Gogna, P. & Qiu, Y. Low-threshold photonic crystal laser. *Appl. Phys. Lett.* **81**, 2680–2682 (2002).
- Park, H.-G. et al. Electrically driven single-cell photonic crystal laser. *Science* **305**, 1444–1447 (2004).
- Altug, H., Englund, D. & Vučković, J. Ultrafast photonic crystal nanocavity laser. *Nat. Phys.* **2**, 484–488 (2006).
- Mao, X.-R., Shao, Z.-K., Luan, H.-Y., Wang, S.-L. & Ma, R.-M. Magic-angle lasers in nanostructured moiré superlattice. *Nat. Nanotechnol.* **16**, 1099–1105 (2021).
- Ouyang, Y.-H., Luan, H.-Y., Zhao, Z.-W., Mao, W.-Z. & Ma, R.-M. Singular dielectric nanolaser with atomic-scale field localization. *Nature* **632**, 287–293 (2024).
- Wang, W. et al. The rich photonic world of plasmonic nanoparticle arrays. *Mater. Today* **21**, 303–314 (2018).
- Zheludev, N. I., Prosvirnin, S. L., Papasimakis, N. & Fedotov, V. A. Lasing spaser. *Nat. Photon.* **2**, 351–354 (2008).
- Zou, S., Janel, N. & Schatz, G. C. Silver nanoparticle array structures that produce remarkably narrow plasmon lineshapes. *J. Chem. Phys.* **120**, 10871–10875 (2004).
- Markel, V. A. Divergence of dipole sums and the nature of non-Lorentzian exponentially narrow resonances in one-dimensional periodic arrays of nanospheres. *J. Phys. B Mol. Opt. Phys.* **38**, L115–L121 (2005).
- Hill, M. T. et al. Lasing in metal-insulator-metal sub-wavelength plasmonic waveguides. *Opt. Express* **17**, 11107–11112 (2009).
- Oulton, R. F. et al. Plasmon lasers at deep subwavelength scale. *Nature* **461**, 629–632 (2009).
- Noginov, M. A. et al. Demonstration of a spaser-based nanolaser. *Nature* **460**, 1110–1112 (2009).
- Bergman, D. J. & Stockman, M. I. Surface plasmon amplification by stimulated emission of radiation: quantum generation of coherent surface plasmons in nanosystems. *Phys. Rev. Lett.* **90**, 027402 (2003).
- Wang, S. et al. Unusual scaling laws for plasmonic nanolasers beyond the diffraction limit. *Nat. Commun.* **8**, 1889 (2017).
- Kravets, V. G., Schedin, F. & Grigorenko, A. N. Extremely narrow plasmon resonances based on diffraction coupling of localized plasmons in arrays of metallic nanoparticles. *Phys. Rev. Lett.* **101**, 087403 (2008).
- Zhou, W. et al. Lasing action in strongly coupled plasmonic nanocavity arrays. *Nat. Nanotechnol.* **8**, 506–511 (2013).
- Chan, G. H., Zhao, J., Hicks, E. M., Schatz, G. C. & Duyne, R. P. V. Plasmonic properties of copper nanoparticles fabricated by nanosphere lithography. *Nano Lett.* **7**, 1947–1952 (2007).
- Freire-Fernández, F., Kataja, M. & van Dijken, S. Surface-plasmon-polariton-driven narrow-linewidth magneto-optics in Ni nanodisk arrays. *Nanophotonics* **9**, 113–121 (2019).
- Freire-Fernández, F., Mansell, R. & Dijken, S. V. Magnetoplasmonic properties of perpendicularly magnetized [Co/Pt]N nanodots. *Phys. Rev. B* **101**, 054416 (2020).
- Rawashdeh, A., Lupa, S., Welch, W. & Yang, A. Sodium surface lattice plasmons. *J. Phys. Chem. C* **125**, 25148–25154 (2021).
- Juarez, X. G. et al. Chiral optical properties of plasmonic kagome lattices. *ACS Photon.* **11**, 673–681 (2024).
- Juarez, X. G. et al. M-point lasing in hexagonal and honeycomb plasmonic lattices. *ACS Photon.* **9**, 52–58 (2022).
- Schokker, A. H. & Koenderink, A. F. Lasing in quasi-periodic and aperiodic plasmon lattices. *Optica* **3**, 686 (2016).
- Yang, A. et al. Real-time tunable lasing from plasmonic nanocavity arrays. *Nat. Commun.* **6**, 1–7 (2015).
- Wang, D. et al. Band-edge engineering for controlled multi-modal nanolasing in plasmonic superlattices. *Nat. Nanotechnol.* **12**, 889–894 (2017).
- Fernandez-Bravo, A. et al. Ultralow-threshold, continuous-wave upconverting lasing from subwavelength plasmons. *Nat. Mater.* **18**, 1172–1176 (2019).
- Schokker, A. H. & Koenderink, A. F. Statistics of randomized plasmonic lattice lasers. *ACS Photon.* **2**, 1289–1297 (2015).
- Hoang, T. B., Akselrod, G. M., Yang, A., Odom, T. W. & Mikkelsen, M. H. Millimeter-scale spatial coherence from a plasmon laser. *Nano Lett.* **17**, 6690–6695 (2017).
- Rekola, H. T., Hakala, T. K. & Törmä, P. One-dimensional plasmonic nanoparticle chain lasers. *ACS Photon.* **5**, 1822–1826 (2018).
- Freire-Fernández, F. et al. Room-temperature polariton lasing from CdSe core-only nanoplatelets. *ACS Nano* **18**, 15177–15184 (2024).
- Ramezani, M. et al. Plasmon-exciton-polariton lasing. *Optica* **4**, 31 (2017).
- Boddeti, A. K. et al. Reducing effective system dimensionality with long-range collective dipole-dipole interactions. *Phys. Rev. Lett.* **132**, 173803 (2024).
- Yadav, R. K. et al. Strongly coupled exciton-surface lattice resonances engineer long-range energy propagation. *Nano Lett.* **20**, 5043–5049 (2020).
- Hakala, T. K. et al. Bose-Einstein condensation in a plasmonic lattice. *Nat. Phys.* **14**, 739–744 (2018).
- Väkeväinen, A. I. et al. Sub-picosecond thermalization dynamics in condensation of strongly coupled lattice plasmons. *Nat. Commun.* **11**, 3139 (2020).
- Kichin, G. et al. Metal-dielectric photonic crystal superlattice: 1D and 2D models and empty lattice approximation. *Phys. B Condens. Matter* **407**, 4037–4042 (2012).
- Cherqui, C., Bourgeois, M. R., Wang, D. & Schatz, G. C. Plasmonic surface lattice resonances: theory and computation. *Acc. Chem. Res.* **52**, 2548–2558 (2019).
- Kelly, K. L., Coronado, E., Zhao, L. L. & Schatz, G. C. The optical properties of metal nanoparticles: the influence of size, shape, and dielectric environment. *J. Phys. Chem. B* **107**, 668–677 (2003).
- Li, R. et al. Hierarchical hybridization in plasmonic honeycomb lattices. *Nano Lett.* **19**, 6435–6441 (2019).
- Zhou, W. & Odom, T. W. Tunable subradiant lattice plasmons by out-of-plane dipolar interactions. *Nat. Nanotechnol.* **6**, 423–427 (2011).
- Dey, D. & Schatz, G. C. Plasmonic surface lattice resonances in nanoparticle arrays. *MRS Bull.* **49**, 421–430 (2024).
- Kravets, V. G., Kabashin, A. V., Barnes, W. L. & Grigorenko, A. N. Plasmonic surface lattice resonances: a review of properties and applications. *Chem. Rev.* **118**, 5912–5951 (2018).
- Pick, A. et al. General theory of spontaneous emission near exceptional points. *Opt. Express* **25**, 12325 (2017).
- Guo, R., Nečada, M., Hakala, T. K., Väkeväinen, A. I. & Törmä, P. Lasing at K points of a honeycomb plasmonic lattice. *Phys. Rev. Lett.* **122**, 1–9 (2019).
- Knudson, M. P. et al. Polarization-dependent lasing behavior from low-symmetry nanocavity arrays. *ACS Nano* **13**, 7435–7441 (2019).
- Guan, J. et al. Plasmonic nanoparticle lattice devices for white-light lasing. *Adv. Mater.* **35**, e2103262 (2023).
- Norris, D. J., Sacra, A., Murray, C. B. & Bawendi, M. G. Measurement of the size dependent hole spectrum in CdSe quantum dots. *Phys. Rev. Lett.* **72**, 2612–2615 (1994).
- Alivisatos, A. P. Semiconductor clusters, nanocrystals, and quantum dots. *Science* **271**, 933–937 (1996).
- Gammon, D., Snow, E. S., Shanabrook, B. V., Katzer, D. S. & Park, D. Homogeneous linewidths in the optical spectrum of a single gallium arsenide quantum dot. *Science* **273**, 87–90 (1996).
- Diroll, B. T. et al. 2D II-VI semiconductor nanoplatelets: from material synthesis to optoelectronic integration. *Chem. Rev.* **123**, 3543–3624 (2023).
- Pietryga, J. M. et al. Spectroscopic and device aspects of nanocrystal quantum dots. *Chem. Rev.* **116**, 10513–10622 (2016).

64. Guan, J. et al. Quantum dot-plasmon lasing with controlled polarization patterns. *ACS Nano* **14**, 3426–3433 (2020).
65. Winkler, J. M. et al. Dual-wavelength lasing in quantum-dot plasmonic lattice lasers. *ACS Nano* **14**, 5223–5232 (2020).
66. Montanarella, F. & Kovalenko, M. V. Three millennia of nanocrystals. *ACS Nano* **16**, 5085–5102 (2022).
67. Kovalenko, M. V., Protesescu, L. & Bodnarchuk, M. I. Properties and potential optoelectronic applications of lead halide perovskite nanocrystals. *Science* **358**, 745–750 (2017).
68. Tan, M. J. H., Wang, Y. & Chan, Y. Solution-based green amplified spontaneous emission from colloidal perovskite nanocrystals exhibiting high stability. *Appl. Phys. Lett.* **114**, 183101 (2019).
69. Jing, L. et al. Aqueous based semiconductor nanocrystals. *Chem. Rev.* **116**, 10623–10730 (2016).
70. Kovalenko, M. V. et al. Prospects of nanoscience with nanocrystals. *ACS Nano* **9**, 1012–1057 (2015).
71. Park, Y.-S., Bae, W. K., Padilha, L. A., Pietryga, J. M. & Klimov, V. I. Effect of the core/shell interface on Auger recombination evaluated by single-quantum-dot spectroscopy. *Nano Lett.* **14**, 396–402 (2014).
72. Tan, M. J. H., Patel, S. K., Chiu, J., Zheng, Z. T. & Odom, T. W. Liquid lasing from solutions of ligand-engineered semiconductor nanocrystals. *J. Chem. Phys.* **160**, 154703 (2024).
73. Wang, Y. et al. Unraveling the ultralow threshold stimulated emission from CdZnS/ZnS quantum dot and enabling high-Q microlasers. *Laser Photon. Rev.* **9**, 507–516 (2015).
74. Haase, M. & Schäfer, H. Upconverting nanoparticles. *Angew. Chem. Int. Ed.* **50**, 5808–5829 (2011).
75. Zhou, B., Shi, B., Jin, D. & Liu, X. Controlling upconversion nanocrystals for emerging applications. *Nat. Nanotechnol.* **10**, 924–936 (2015).
76. Xie, Y.-Y. et al. Metasurface-integrated vertical cavity surface-emitting lasers for programmable directional lasing emissions. *Nat. Nanotechnol.* **15**, 125–130 (2020).
77. Trinh, Q. T. et al. Coexistence of surface lattice resonances and bound states in the continuum in a plasmonic lattice. *Opt. Lett.* **47**, 1510 (2022).
78. Adams, M. J. *An Introduction to Optical Waveguides* (Wiley, 1981).
79. Park, J.-E. et al. Polariton dynamics in two-dimensional Ruddlesden–Popper perovskites strongly coupled with plasmonic lattices. *ACS Nano* **16**, 3917–3925 (2022).
80. Vicidomini, G., Bianchini, P. & Diaspro, A. STED super-resolved microscopy. *Nat. Methods* **15**, 173–182 (2018).
81. Gómez-Viloria, I. et al. On-axis optical trapping with vortex beams: the role of the multipolar decomposition. *ACS Photon.* **11**, 626–633 (2024).
82. Tan, M. J. H., Freire-Fernández, F. & Odom, T. W. Symmetry-guided engineering of polarization by 2D moiré metasurfaces. *ACS Nano* **18**, 23181–23188 (2024).
83. Zhang, H. et al. Generation of orbital angular momentum modes using fiber systems. *Appl. Sci.* **9**, 1033 (2019).
84. Wang, Z. et al. Transmission and generation of orbital angular momentum modes in optical fibers. *Photonics* **8**, 246 (2021).
85. Jain, P. K., Huang, W. & El-Sayed, M. A. On the universal scaling behavior of the distance decay of plasmon coupling in metal nanoparticle pairs: a plasmon ruler equation. *Nano Lett.* **7**, 2080–2088 (2007).
86. Salerno, G. et al. Loss-driven topological transitions in lasing. *Phys. Rev. Lett.* **129**, 173901 (2022).
87. Wang, D., Yang, A., Hryn, A. J., Schatz, G. C. & Odom, T. W. Superlattice plasmons in hierarchical Au nanoparticle arrays. *ACS Photon.* **2**, 1789–1794 (2015).
88. Heilmann, R., Arjas, K., Hakala, T. K. & Törmä, P. Multimode lasing in supercell plasmonic nanoparticle arrays. *ACS Photon.* **10**, 3955–3962 (2023).
89. Fasanelli, F. M., Freire-Fernández, F. & Odom, T. W. Symmetry-determined lasing from incommensurate moiré nanoparticle lattices. *Adv. Opt. Mater.* **12**, 2400797 (2024).
90. Freire-Fernández, F. et al. Quasi-random multimetallic nanoparticle arrays. *ACS Nano* **17**, 21905–21911 (2023).
91. McPeak, K. M. et al. Plasmonic films can easily be better: rules and recipes. *ACS Photon.* **2**, 326–333 (2015).
92. Wang, R., Peng, J., Qiu, F., Yang, Y. & Xie, Z. Simultaneous blue, green, and red emission from diblock copolymer micellar films: a new approach to white-light emission. *Chem. Commun.* **28**, 6723–6725 (2009).
93. Bailey, S. T. et al. Optimized excitation energy transfer in a three-dye luminescent solar concentrator. *Sol. Energy Mater. Sol. Cells* **91**, 67–75 (2007).
94. Wang, R., Peng, J., Qiu, F. & Yang, Y. Enhanced white-light emission from multiple fluorophores encapsulated in a single layer of diblock copolymer micelles. *Chem. Commun.* **47**, 2787 (2011).
95. Deng, S. et al. Ultranarrow plasmon resonances from annealed nanoparticle lattices. *Proc. Natl Acad. Sci. USA* **117**, 23380–23384 (2020).
96. Deng, S. et al. Interfacial engineering of plasmonic nanoparticle metasurfaces. *Proc. Natl Acad. Sci. USA* **119**, e2202621119 (2022).
97. Freire-Fernández, F. et al. Magnetic on–off switching of a plasmonic laser. *Nat. Photon.* **16**, 27–32 (2022).
98. Kataja, M., Pourjamal, S. & van Dijken, S. Magnetic circular dichroism of non-local surface lattice resonances in magnetic nanoparticle arrays. *Opt. Express* **24**, 3562 (2016).
99. Chen, H.-Y. et al. Self-assembled plasmonic nanojunctions mediated by host–guest interaction for ultrasensitive dual-mode detection of cholesterol. *ACS Sens.* **8**, 388–396 (2023).
100. Acharya, A. et al. Ultrathin covalent organic overlayers on metal nanocrystals for highly selective plasmonic photocatalysis. *Nat. Commun.* **14**, 7667 (2023).
101. Byrnes, T., Kim, N. Y. & Yamamoto, Y. Exciton-polariton condensates. *Nat. Phys.* **10**, 803–813 (2014).
102. Ramezani, M., Berghuis, M. & Rivas, J. G. Strong light–matter coupling and exciton-polariton condensation in lattices of plasmonic nanoparticles [Invited]. *J. Opt. Soc. Am. B* **36**, E88 (2019).
103. Törmä, P. & Barnes, W. L. Strong coupling between surface plasmon polaritons and emitters: a review. *Rep. Prog. Phys.* **78**, 013901 (2015).
104. Giorgi, M. D. et al. Interaction and coherence of a plasmon-exciton polariton condensate. *ACS Photon.* **5**, 3666–3672 (2018).
105. Ramezani, M., Le-Van, Q., Halpin, A. & Rivas, J. G. Nonlinear emission of molecular ensembles strongly coupled to plasmonic lattices with structural imperfections. *Phys. Rev. Lett.* **121**, 243904 (2018).
106. Christopoulos, S. et al. Room-temperature polariton lasing in semiconductor microcavities. *Phys. Rev. Lett.* **98**, 126405 (2007).
107. Watkins, N. E. et al. Surface normal lasing from CdSe nanoplatelets coupled to aluminum plasmonic nanoparticle lattices. *J. Phys. Chem. C* **125**, 19874–19879 (2021).
108. Deng, H., Weihs, G., Snoke, D., Bloch, J. & Yamamoto, Y. Polariton lasing vs. photon lasing in a semiconductor microcavity. *Proc. Natl Acad. Sci. USA* **100**, 15318–15323 (2003).
109. Koenderink, A. F. Plasmon nanocavity array lasers: cooperating over losses and competing for gain. *ACS Nano* **13**, 7377–7382 (2019).
110. Guan, J. et al. Far-field coupling between moiré photonic lattices. *Nat. Nanotechnol.* **18**, 514–520 (2023).
111. Jaynes, E. T. & Cummings, F. W. Comparison of quantum and semiclassical radiation theories with application to the beam maser. *Proc. IEEE* **51**, 89–109 (1963).
112. Arjas, K., Taskinen, J. M., Heilmann, R., Salerno, G. & Törmä, P. High topological charge lasing in quasicrystals. *Nat. Commun.* **15**, 9544 (2024).
113. Che, Z. et al. Polarization singularities of photonic quasicrystals in momentum space. *Phys. Rev. Lett.* **127**, 043901 (2021).
114. Harari, G. et al. Topological insulator laser: theory. *Science* **359**, eaar4003 (2018).
115. Liu, Y. G. N. et al. Complex skin modes in non-Hermitian coupled laser arrays. *Light Sci. Appl.* **11**, 336 (2022).
116. Heilmann, R., Salerno, G., Cuerda, J., Hakala, T. K. & Törmä, P. Quasi-BIC mode lasing in a quadrumer plasmonic lattice. *ACS Photon.* **9**, 224–232 (2022).
117. Greybush, N. J., Pacheco-Peña, V., Engheta, N., Murray, C. B. & Kagan, C. R. Plasmonic optical and chiroptical response of self-assembled Au nanorod equilateral trimers. *ACS Nano* **13**, acsnano.8b07619 (2019).
118. Fan, J. A. et al. Self-assembled plasmonic nanoparticle clusters. *Science* **328**, 1135–1138 (2010).
119. Yegilyurt, A. T. M. et al. Unidirectional meta-emitters based on the Kerker condition assembled by DNA origami. *ACS Nano* **17**, 19189–19196 (2023).
120. Schuknecht, F., Kotłaj, K., Steinberger, M., Liedl, T. & Lohmueller, T. Accessible hotspots for single-protein SERS in DNA-origami assembled gold nanorod dimers with tip-to-tip alignment. *Nat. Commun.* **14**, 7192 (2023).
121. Cerdán, L., Zundel, L. & Manjavacas, A. Chiral lattice resonances in 2.5-dimensional periodic arrays with achiral unit cells. *ACS Photon.* **10**, 1925–1935 (2023).
122. Zhao, Y., Belkin, M. A. & Alù, A. Twisted optical metamaterials for planarized ultrathin broadband circular polarizers. *Nat. Commun.* **3**, 870 (2012).
123. Fradkin, I. M., Dyakov, S. A. & Gippius, N. A. Thickness-independent narrow resonance in a stack of plasmonic lattices. *Phys. Rev. Appl.* **14**, 1–9 (2020).
124. Becerril, D., Pirruccio, G. & Noguez, C. Optical band engineering via vertical stacking of honeycomb plasmonic lattices. *Phys. Rev. B* **103**, 1–12 (2021).
125. Lin, Q.-Y. et al. Building superlattices from individual nanoparticles via template-confined DNA-mediated assembly. *Science* **359**, 669–672 (2018).
126. Rodrigues, S. P., Lan, S., Kang, L., Cui, Y. & Cai, W. Nonlinear imaging and spectroscopy of chiral metamaterials. *Adv. Mater.* **26**, 6157–6162 (2014).
127. Shafiei, F. et al. A subwavelength plasmonic metamolecule exhibiting magnetic-based optical Fano resonance. *Nat. Nanotechnol.* **8**, 95–99 (2013).
128. Park, C. et al. 12" wafer-scale mass-manufactured metal-insulator-metal reflective meta-holograms by nanotransfer printing. *ACS Appl. Mater. Interfaces* **17**, 3749–3756 (2025).
129. Yang, F. et al. Fabrication of centimeter-scale plasmonic nanoparticle arrays with ultranarrow surface lattice resonances. *ACS Nano* **17**, 725–734 (2023).
130. Shen, X. et al. Engineering the electrostatic interactions between oppositely charged polymer-grafted nanoparticles for constructing colloid molecules on substrates. *ACS Nano* **18**, 20999–21008 (2024).
131. Ye, S. et al. Centimeter-scale superlattices of three-dimensionally orientated plasmonic dimers with highly tunable collective properties. *ACS Nano* **16**, 4609–4618 (2022).
132. Cho, N. H., Jia, J., Park, S.-M., Wen, X. & Odom, T. W. Templated synthesis of mono- and bimetallic nanogap dimer arrays. *ACS Nano* **19**, 8966–8973 (2025).
133. Juliano Martins, R. et al. Metasurface-enhanced light detection and ranging technology. *Nat. Commun.* **13**, 5724 (2022).
134. Hu, J. et al. Diffractive optical computing in free space. *Nat. Commun.* **15**, 1525 (2024).
135. Kim, H. et al. Optical metasurfaces for biomedical imaging and sensing. *ACS Nano* **19**, 3085–3114 (2025).
136. Zakharko, Y. et al. Surface lattice resonances for enhanced and directional electroluminescence at high current densities. *ACS Photon.* **3**, 2225–2230 (2016).
137. Ahn, N. et al. Electrically driven amplified spontaneous emission from colloidal quantum dots. *Nature* **617**, 79–85 (2023).

138. Ahn, N. et al. Optically excited lasing in a cavity-based, high-current-density quantum dot electroluminescent device. *Adv. Mater.* **35**, e2206613 (2023).
139. Ahn, N., Livache, C., Pinchetti, V. & Klimov, V. I. Colloidal semiconductor nanocrystal lasers and laser diodes. *Chem. Rev.* **123**, 8251–8296 (2023).
140. Sinai, N. G. et al. Electrochemical control of strong coupling of CdSe exciton-polaritons in plasmonic cavities. *Nano Lett.* **24**, 7491–7498 (2024).
141. Fischer, A. et al. Surface lattice resonance lasers with epitaxial InP gain medium. *ACS Photon.* **11**, 4316–4322 (2024).
142. Wang, Y. et al. Stable, high-performance sodium-based plasmonic devices in the near infrared. *Nature* **581**, 401–405 (2020).

Acknowledgements

This work was supported by the Vannevar Bush Faculty Fellowship from the US Department of Defense (DOD N00014-17-1-3023), the National Science Foundation (NSF) under DMR-2207215 and the Office of Naval Research (ONR N00014-21-1-2289). M.J.H.T. and S.-M.P. gratefully acknowledge support from the Ryan Fellowship and the International Institute of Nanotechnology at Northwestern University.

Author contributions

F.F.-F., S.-M.P. and M.J.H.T. researched data for the article. F.F.-F., S.-M.P. and T.W.O. led the drafting of the manuscript. All authors contributed substantially to the discussion of the

content and drafting of the article. F.F.-F., S.-M.P. and T.W.O. reviewed and/or edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

Additional information

Peer review information *Nature Reviews Materials* thanks Shangjr Gwo and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© Springer Nature Limited 2025